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/ **Practice Exam - 1**

Correct Answer

Next Q

Practice Exam - 1

Question 99 of 117



The patient is a 62-year-old female with chronic stage C heart failure with reduced ejection fraction secondary to nonischemic cardiomyopathy, diabetes, hypertension and CKD. She established care with a cardiologist and her GDMT was adjusted. She is maintained on sacubitril-valsartan 97-103 mg twice daily, carvedilol 6.25 mg twice daily, spironolactone 25 mg once daily empagliflozin 10 mg daily, furosemide 20 mg daily and metformin 500mg BID.

She has remained stable on this regimen for the prior year and now presents to the office complaining of worsening dyspnea on exertion (previously able to walk a mile and now develops dyspnea with 2 blocks), orthopnea and abdominal bloating. On examination, her blood pressure is 115/72 mmHg (stable from prior), her heart rate is 82 bpm and she is saturating 98% on room air. Her JVP is 12 cm of h20 with a positive HJR. She has no signs of LE edema or ascites. Blood work is notable for a Creatinine of 1.6 mg/dl and a potassium of 4.7 mEq/L.

Which would be the appropriate next management strategy?

A. Add metolazone 5 mg daily



B. Add acetazolamide 500 mg daily

C. Increase Spironolactone to 50 mg daily



D. Increase furosemide to 40 mg one to twice daily

Commentary

References

Testing point: The board-certified AHFTC specialist should recognize that patients not responding to low dose loop diuretics should be treated with escalation of loop diuretic dose prior to addition of thiazide diuretics.

Correct Answer: D. Increase furosemide to 40mg once to twice daily

Rationale: The patient is presenting with fluid retention and decompensation of her chronic heart failure and needs decongestion. An approach of first increasing loop diuretic doses is supported in the 2023 ACC/AHA/HFSA HF guidelines which state that “addition of

thiazide diuretics to loop diuretic should be reserved for patients who do not respond to moderate or high-dose loop diuretics to minimize electrolyte abnormalities". Furosemide 20 mg is a low-dose and should thus be titrated for decongestion as an initial strategy in this patient (correct answer D) . While adding a thiazide diuretic (Answer A) does increase diuresis by inhibiting compensatory distal tubular sodium absorption, it increases the risk of electrolyte abnormalities namely hypokalemia and hyponatremia as well as worsening renal function. While RCT data guiding diuretic choices is limited, the DOSE (Diuretic Optimization Strategies Evaluation) trial supports the safety of high-dose diuretics.

Answer B: Acetazolamide has recently been shown in the ADVOR trial to increase natriuresis in patients with acute decompensated heart failure and to be associated with a decrease in the rates of overt congestion compared to intravenous loop diuretic alone. Outpatients on low dose oral loop diuretic were not included in this trial.

Answer C: While sequential nephron blockage with adjunctive non loop diuretic may be of benefit, such as adding or increasing acetazolamide or spironolactone, the first step should be to increase loop diuretics to at least moderate doses.

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